

Srujan Vaddiparthi

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SUMMARY

Data Scientist with experience building applied machine learning and analytics systems across industrial engineering and e-commerce environments. Skilled in developing interpretable models, stability-aware ensemble pipelines, and decision-support tools that translate complex data into actionable insights for engineers, managers, and business teams. Interested in solving real-world problems where machine learning must operate under practical constraints.

WORK EXPERIENCE

[ITT Goulds Pumps](#), Seneca Falls, NY

Data Scientist (R&D - Hydraulic Engineering)

Jan 2025 - Dec 2025

- Replaced legacy heuristic models with a domain-informed ensemble ML system, improving holdout regression performance (**R-squared from 0.33 to 0.68**) while achieving balanced classification (**precision/recall/F1 = 0.80**) through statistically validated model evaluation.
- Productized the ensemble ML system into a deployable **decision-support and visualization application** (Python, Tkinter, PyInstaller), now **actively used** by hydraulic engineers and leadership to accelerate pump design reviews; translated model outputs into operational engineering decisions, authored user documentation, and delivered executive-facing demonstrations.
- Integrated an engineer-facing **explainability and sensitivity analysis toolkit** into the deployed application using SHAP and domain-constrained ICE, enabling transparent communication of model drivers, supporting stakeholder trust, and reducing **weeks-to-months** of CFD-driven iteration to **minutes-scale ML-guided screening (<12 minutes for 20 designs)**.
- Designed and implemented **normalized CV+ conformal prediction** on a 100-model ensemble, generating calibrated prediction intervals (80–95% coverage) to support **uncertainty-aware engineering decisions**.
- Developed a domain-informed feature selection framework that reduced inputs from **40 to 5-8 variables** per target using statistical testing, permutation importance, and expert validation, improving **model stability and interpretability** on a small, imbalanced dataset.
- Developed a stability-aware training strategy using **Optuna**, stratified bootstrapping, and cross-validation to mitigate overfitting and improve model reliability under small-sample constraints (**96-row dataset**).
- Reverse engineered a legacy ML tool for pump performance prediction, identifying data inconsistencies and replacing heuristic confidence metrics with statistically grounded uncertainty estimation (**20% early R-squared improvement**).
- Selected as **1 of 5 panelists (from 81 interns)** for ITT Inc.'s Voices of the Next Generation World Youth Skills Day discussion, sharing experiences on applied ML in industrial R&D with ~100 employees.

[Rochester Institute of Technology](#), Rochester, NY

Teaching Assistant

Aug 2024 - Present

- Mentored and evaluated 61 students across three semesters in Python and Java courses, reinforcing algorithmic reasoning, debugging discipline, and structured problem decomposition.

[Udaan – Business eCommerce](#), Bengaluru, India

Business Intelligence Analyst

Jan 2023 - Jun 2023

- Engineered a quantity-based ladder pricing framework by modeling historical order distributions and demand sensitivity, increasing average order value by **70%** while preserving inventory balance across categories.
- Designed and evaluated controlled A/B pricing experiments under inventory and operational constraints, performing post-test hypothesis testing to translate experimental outcomes into statistically informed rollout decisions.
- Formalized purchase-order approval logic by integrating daily inventory run-rate features into automated decision rules, reducing approval latency by **80%** and improving system reliability under imperfect data conditions.
- Built structured analytical views to support pricing and advertising optimization, decomposing order funnel metrics and validating quantitative findings with field observations to reduce decision noise.

[BITS Pilani Mechanical Engineering Department](#), Hyderabad, India

Research Assistant

Jul 2022 - Dec 2022

- Designed and built a multi-axis CNC gantry robot for automated planting and weed removal, integrating motion control, sensors, and modular tool attachments within a constrained physical system.
- Led system calibration and debugging across mechanical and embedded control layers, validating repeatable positioning under real-world tolerances and documenting experimental performance in a formal technical report.
- Implemented [proof-of-concept](#) YOLOv7 object detection experiments on agricultural imagery, evaluating perception feasibility under lighting and field variability constraints.

PROJECTS

Weed Species Classification (Transfer Learning & Interpretability), Rochester, New York

Course Project - Applications of Machine Learning for Remote Sensing

Oct 2025 - Dec 2025

- Built a fully reproducible PyTorch training pipeline for the DeepWeeds dataset, implementing stratified splits, controlled augmentation, and deterministic experimentation to ensure stable performance estimation.
- Fine-tuned an ImageNet-pretrained ResNet-50 under small-data constraints, achieving **94% test accuracy** while evaluating performance stability across multiple random seeds to guard against split sensitivity.
- Conducted class-wise error decomposition using confusion matrices and per-class metrics, identifying systematic misclassification patterns among visually similar species and background-heavy samples.
- Applied **Grad-CAM** interpretability to assess feature attribution validity and verify that performance gains reflected biologically meaningful structures rather than spurious background correlations.

Synthetic Industrial Surface Defect Generation Under Low-Data Constraints (Ongoing Research), Rochester, New York

Capstone Project

Aug 2024 - Present

- Developed and evaluated **conditional DCGAN pipelines** in TensorFlow/Keras for defect-faithful industrial surface image generation under scarce, class-imbalanced data conditions across normal, scratch, and spot classes.
- Built a staged low-data experimentation framework spanning 3-class generation, binary decomposition, oversampling, stratified batch sampling, randomized defect-specific enhancement, and mild geometric augmentation.
- Refactored training into a stable **tf.data + GradientTape** workflow on RIT Research Computing infrastructure; evaluated outputs using realism, class consistency, and diversity, finding **spot defects more learnable than scratches** while diversity remained limited.

DineNY – Restaurant Search Platform (FastAPI, MongoDB), Rochester, New York

Course Project - Non Relational Data Management

Feb 2026 - Present

- Processed and filtered a **~380K row raw dataset**, engineering a consolidated search_blob field from multiple text attributes and a standardized schema to optimize full-text indexing and enable efficient downstream querying.
- Designed a **MongoDB** document-oriented **data model** and implemented aggregation pipelines for multi-criteria filtering and ranking, with 2dsphere geospatial indexes enabling proximity-based restaurant discovery.
- Built a **FastAPI**-backed RESTful **search layer** with composable query parameters (text, rating, geospatial radius), dynamically constructing MongoDB aggregation pipelines per request to support flexible multi-dimensional querying.
- Architected an end-to-end system spanning **data ingestion** → **transformation** → **NoSQL modeling** → **API layer**, transforming raw semi-structured data into a queryable, application-ready product comparable to production location-based search platforms.

VOLUNTEERING

Nirmaan NGO, BITS Pilani, Hyderabad, India

Volunteer Tutor

Feb 2021 - Dec 2022

- Tutored middle-school students in math and science as part of Project Parishkaar.

SKILLS

- **Programming Languages:** Python, SQL, Java
- **Machine Learning:** Regression, Classification, Ensemble Methods, Generative Modeling, Hyperparameter Optimization, Model Evaluation, Feature Engineering
- **Statistical Methods:** Statistical Modeling, Hypothesis Testing, Experimental Design, ANOVA, Kruskal-Wallis, Regression Diagnostics, Bootstrapping, Uncertainty Quantification
- **Industrial and Time-Series Analytics:** Anomaly Detection, Reliability Modeling, Time-Series Feature Extraction, FFT Analysis, Root Cause Analysis
- **Experimentation and Causal Analysis:** A/B Testing, Multivariate Experimentation, Statistical Inference
- **ML and Data Libraries:** scikit-learn, TensorFlow, Keras, PyTorch, Torchvision, NumPy, Pandas
- **Data Engineering and Systems:** Data Pipeline Development, Data Ingestion (CSV/XML), ETL Workflows, Data Validation, Data Quality Monitoring, FastAPI, MongoDB (NoSQL), AWS S3, Google BigQuery
- **Visualization and Data Products:** Tableau, Amazon QuickSight, Matplotlib, Seaborn
- **Tools:** Git, Jupyter, Conda, Joblib
- **Hardware and Embedded:** Arduino, Sensors, Motion Control Systems

EDUCATION

Rochester Institute of Technology (RIT), Rochester, NY

Master of Science in Data Science

Aug 2023 - Aug 2026

Birla Institute of Technology and Sciences - Pilani (BITS), Hyderabad, India

Bachelor of Engineering in Mechanical Engineering, Minor - Robotics and Automation

Aug 2019 - May 2023